

**8** Sehuencas water frogs are endemic to Bolivia.

It was once believed that this frog was extinct in the wild.

This was due to human activity and a disease caused by a fungus.

- (a) Suggest how human activity could cause the extinction of the Sehuencas water frog in the wild.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



- (b) In 2008, there was only one Sehuencas water frog in captivity.

In 2018, five Sehuencas water frogs were discovered in a Bolivian mountain forest. Three of these frogs were male.

Scientists suggested that these five frogs were resistant to the fungal disease.

Explain how resistance to this fungus could develop in a population of Sehuencas water frogs.

(4)

- (c) One way of studying the genetic diversity within a species is to calculate the heterozygosity index.

Describe how the scientists could calculate the heterozygosity index for a population of Sehuencas water frogs.

(2)



\*(d) Describe the role of zoos in the conservation of endangered Sehuencas water frogs.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 = 15 marks)

**TOTAL FOR PAPER = 80 MARKS**

